

FedREST-HIDS

Code accompanying "**FedREST-HIDS: Split-Federated Intrusion Detection for Resource-Heterogeneous Industrial IoT**" (Vyas, Hwang & Lin), submitted to *Computer Communications*.

FedREST-HIDS partitions intrusion detection across three tiers. Resource-rich devices train complete local models and take part in federated learning. Resource-constrained devices execute only the initial feature-extraction layer and send clipped, noised activations to an edge server that runs the remaining layers. Federated transfer learning then aligns the edge models globally, and a trust-aware aggregation rule bounds the influence of any single client. Privacy is enforced locally under Rényi differential privacy.

Verify the article's numbers in one second

Every privacy and communication figure printed in the article is checked against this implementation. No dataset, no GPU.

```
bash
```

```
python scripts/verify_against_paper.py
```

```
Privacy accounting -- Section 3.2.2, Section 5.2, Table 4
[PASS] sigma, WUSTL headline (C=1, a=5, eps=0.5)      got 2.2361   article
2.2360
[PASS] eps_DP at delta=1e-5, alpha=5                  got 27.8803  article
27.8800
...
24/24 checks passed
```

Layout

```
fedrest-hids/
├── src/fedrest/           reference implementation
│   ├── privacy.py        Rényi DP: noise scale, composition, RDP⇒DP, allocation
│   ├── fedtrust.py        trust scoring and softmax-weighted aggregation
│   ├── comms.py           communication cost model (Section 6.5)
│   ├── models.py          CNN-HIDS, client-side and edge-side split models
│   ├── training.py        Algorithms 1-4
│   ├── legacy.py          the alternative noise expression used in some notebooks
│   └── poisoning.py       label-flip and sign-flip attacks
├── notebooks/            the 16 notebooks behind the reported results
│   └── MANIFEST.md        which notebook produced which table or figure
├── scripts/
│   ├── verify_against_paper.py 24 checks against the article
│   ├── audit_configs.py       each notebook's config vs the article
│   └── make_figure2.py         regenerates Figure 2 from the training log
├── results/               verification output, regenerated figure
└── docs/KNOWN_ISSUES.md   where code and article agree, and where they don't
```

The notebooks are Colab exports preserved **as run**. Refactoring them would break their correspondence with the article, so `src/fedrest/` holds a cleaned reference version of the mechanisms they each reimplement.

Read `docs/KNOWN_ISSUES.md` before using this code.

Install

```
bash

git clone https://github.com/<user>/fedrest-hids.git
cd fedrest-hids
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
```

Python 3.10+. Both PyTorch and TensorFlow are listed because twelve notebooks use the former and three the latter.

Datasets

Not redistributed. Obtain them from the original sources:

Dataset	Source
WUSTL-IIoT-2021	https://www.cse.wustl.edu/~jain/iiot2/index.html
ToN-IoT	https://research.unsw.edu.au/projects/toniot-datasets
SCVIC-APT-2021	https://ieee-dataport.org/documents/scvic-apt-2021

Place the CSVs under `(data/)` (git-ignored) and adjust the path at the top of whichever notebook you run — see `(docs/DATA_PATHS.md)`. The notebooks were written for Colab and mount Google Drive; those cells need editing for local use.

Using the library

```
python

from fedrest import LocalRenyiDP, FedTrust, CommsProfile, build_split_pair

dp = LocalRenyiDP(clip_norm=1.0, alpha=5.0, epsilon=0.5)
noised = dp.privatize.gradient()
print(dp.report())
# rounds=1 alpha=5.0 eps_per_round=0.5 sigma=2.2361 ...

client, edge = build_split_pair(n_features=41, n_classes=2)
print(CommsProfile(n_samples=28_000,
                  activation_dim=client.activation_dim,
                  n_parameters=100_000).summary())
```

Reproducibility

Stated plainly, so nobody is misled:

- **The federation is simulated.** Clients run sequentially in one process. No wall-clock latency or energy figure is reported anywhere.
- **Headline results are single runs at seed 42.** Tables 6, 8 and 9 report three-seed means with standard deviations; the headline numbers do not.
- **Exact reproduction needs the original Colab environment.** GPU model, library versions and CUDA version affect the third decimal place.
- **The noise-scale expression differs between notebooks.** See KNOWN_ISSUES §2.

Citing

See `(CITATION.cff)`. Please cite both the software and the article.

Licence

MIT — see LICENSE. The datasets carry their own licences.